

3 BATTERY MANAGEMENT SYSTEM IN ELECTRIC VEHICLES

In India, high cost of electric vehicles (EVs) is mainly attributed to their batteries still being imported, which incurs heavy expenses. The government has come up with a number of policies aimed towards boosting EV manufacturing in India

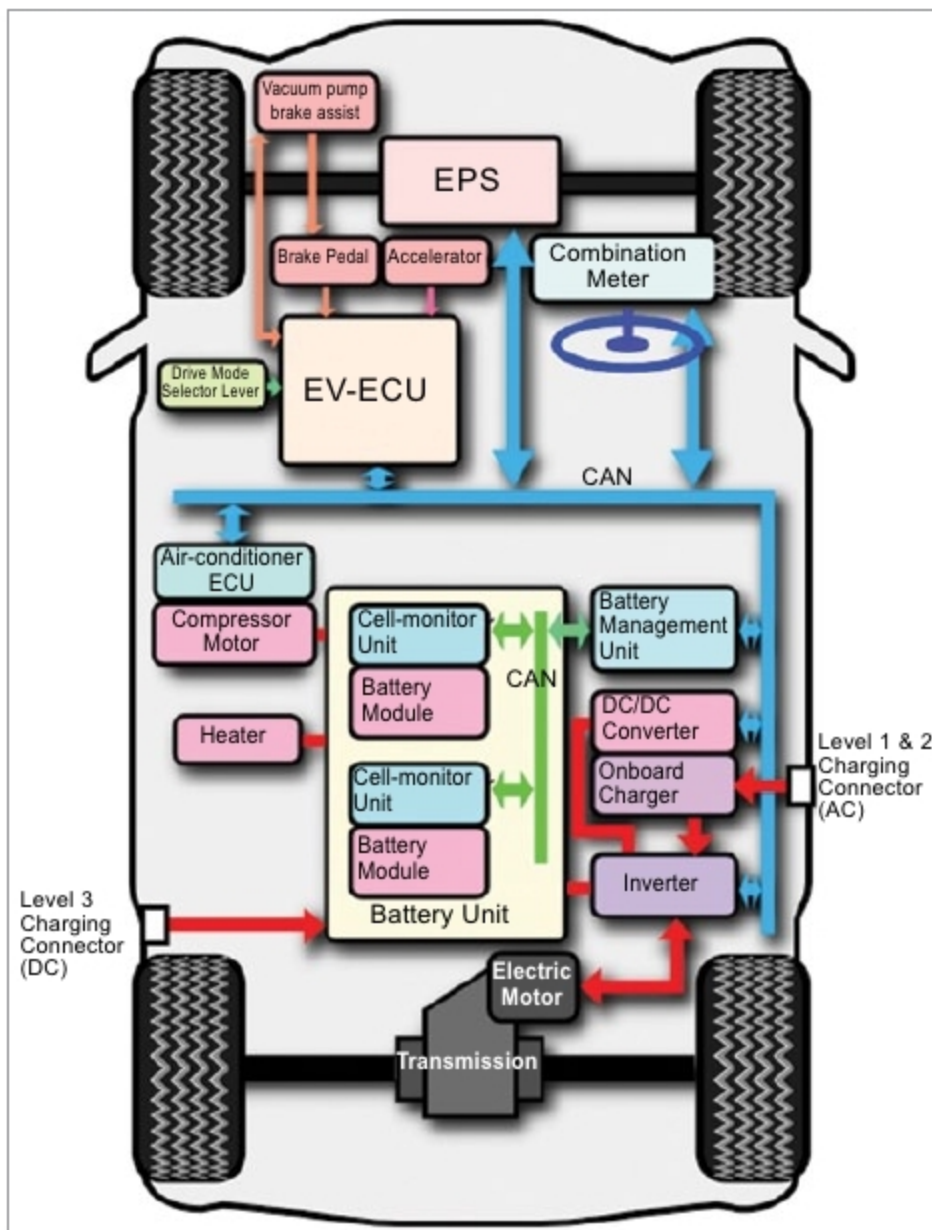
Electric vehicles (EVs) are gradually being adopted as the mainstream mode of transport across the world. An EV consists of a battery pack, motor, power converter and mechanical transmission system. Most EVs use lithium-ion batteries as the source of power due to such factors as high energy density, low discharge rate and long lifespan.

Battery failure can compromise the safety of drivers and passengers. It is therefore necessary to monitor every cell in the battery pack and measure crucial information precisely. This calls for a battery management system (BMS) dedicated entirely to protect the battery and prolong its life.

BMS in an EV comprises multiple types of sensors, actuators and controllers. It acts as the transmission part of the electronic control unit (ECU) to allow continuous communication between the battery pack and the ECU through standard protocols like controller area network (CAN).

The BMS needs to be reliable and highly efficient with minimum accuracy of 1mV to 2mV. It is powered continuously from the EV's battery itself. Hence, it must consume less power but have a high processing speed, so that load on the battery is not too high. Also, a large memory is required to store monitored data.

- Primary function of BMS is to protect every cell. Performance of a lithium-ion battery varies with age, hysteresis and temperature variations. To prevent this, cell balancing and thermal management are essential, as



explained below.

A battery should not be recharged or discharged above or below its permissible limits, respectively, to avoid damage. If voltage of even one cell is unequal to others, efficiency will decrease. Voltage and current measurement units measure voltage and current of every cell inside a battery pack. Battery life also gets improved if a deep discharge is avoided. BMS has a dedicated lithium-ion protector to ensure proper charging and discharging.

If the battery is exposed to extremely high or low temperatures, it will negatively impact performance. For a lithium-ion battery, measured cell temperatures should lie between 25°C and 40°C. The preferred thermal

system for battery cooling is liquid cooling. The temperature control unit measures temperature of the battery and coolant.

- Another important function performed by BMS is energy management. This includes state of available power (SOP), state of charge (SOC), state of life (SOL) and state of health (SOH) estimation.

SOC is used to know the remaining power of the battery. The balancing control unit equalises SOC of the cells in battery packs.

SOH indicates the age and expected life cycle of the pack. The digital output displays balancing work indicator, failure alarm along with SOC and SOH.

Battery swapping. To eliminate problems such as battery degradation by using fast chargers, or to deal with battery failures, battery swapping stations could be a cost-effective

solution. Charge level of batteries may be determined using the Internet of Things (IoT). At the station, depleted batteries can be swapped with fully-charged, low-weighting ones within no time.

Battery architecture of the EV must be such that battery swapping is easy. This also reduces pressure on local power distribution companies.

Old batteries that are replaced can be used as a cheap alternative to home energy-storage units as these still have the capacity to hold a charge. For example, Nissan has partnered with Eaton to convert old batteries into home electricity-storage units.

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